

SMART IRRIGATION DIGITAL TWIN USING MATLAB, SIMULINK, AND IoT

1. Introduction

Agriculture consumes a major portion of global freshwater resources, and inefficient irrigation practices lead to significant water wastage. Traditional irrigation systems operate on fixed schedules or simple threshold logic, ignoring plant stress, evapotranspiration, and environmental dynamics. With climate change and increasing water scarcity, there is a strong need for intelligent, adaptive irrigation systems.

This project presents a Smart Irrigation System based on a Digital Twin approach, integrating MATLAB, Simulink, IoT sensors, ThingSpeak cloud, and ESP32 hardware. The system continuously monitors soil moisture, temperature, and humidity, estimates internal soil and plant states using a digital twin model, and automatically controls irrigation in real time.

2. Objectives

The main objectives of this project are:

- To design a digital twin model of soil moisture and plant stress
- To collect real-time sensor data using low-cost IoT hardware
- To analyze sensor data using MATLAB models
- To implement closed-loop irrigation control
- To reduce water usage while maintaining safe soil moisture levels
- To validate the system using MATLAB simulation, Simulink modeling, and live IoT data

3. System Overview

The proposed system consists of three tightly integrated layers:

1. Physical Layer
 - ESP32 microcontroller
 - Soil moisture sensor (simulated using potentiometer in Wokwi)
 - DHT22 temperature and humidity sensor
 - Relay module for irrigation valve control
2. Digital Twin Layer (MATLAB & Simulink)
 - Soil moisture dynamic model
 - Evapotranspiration model
 - Plant stress model
 - State estimator (digital twin)
 - Intelligent irrigation controller
3. Cloud & Communication Layer
 - ThingSpeak for data logging and command exchange

- MATLAB Online for real-time data processing

4. Hardware Components Used

- ESP32 Development Board
- Soil Moisture Sensor (Potentiometer used for simulation)
- DHT22 Temperature and Humidity Sensor
- Relay Module
- Power Supply (12V adapter / Li-ion batteries with buck converter)

5. Software Tools Used

- MATLAB Online
- Simulink
- ThingSpeak IoT Platform
- Arduino IDE / Wokwi Simulator
- GitHub for version control

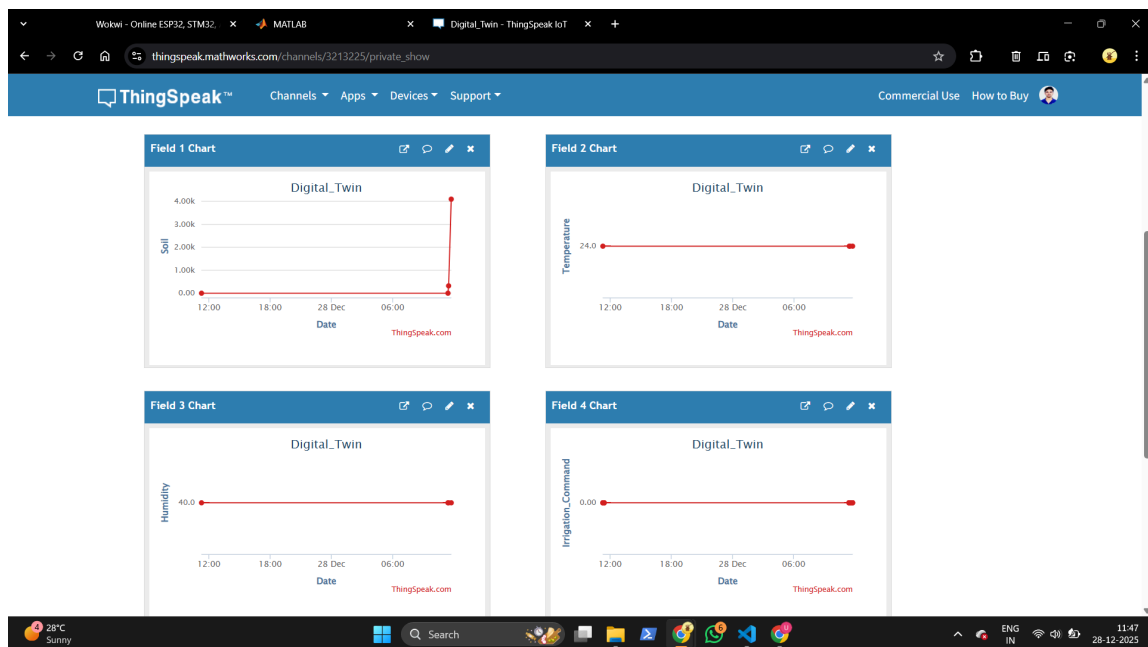
6. Methodology

6.1 Data Acquisition

The ESP32 continuously reads:

- Soil moisture (ADC value 0–4095)
- Temperature (°C)
- Humidity (%)

The sensor data is filtered and uploaded to ThingSpeak every 20 seconds.

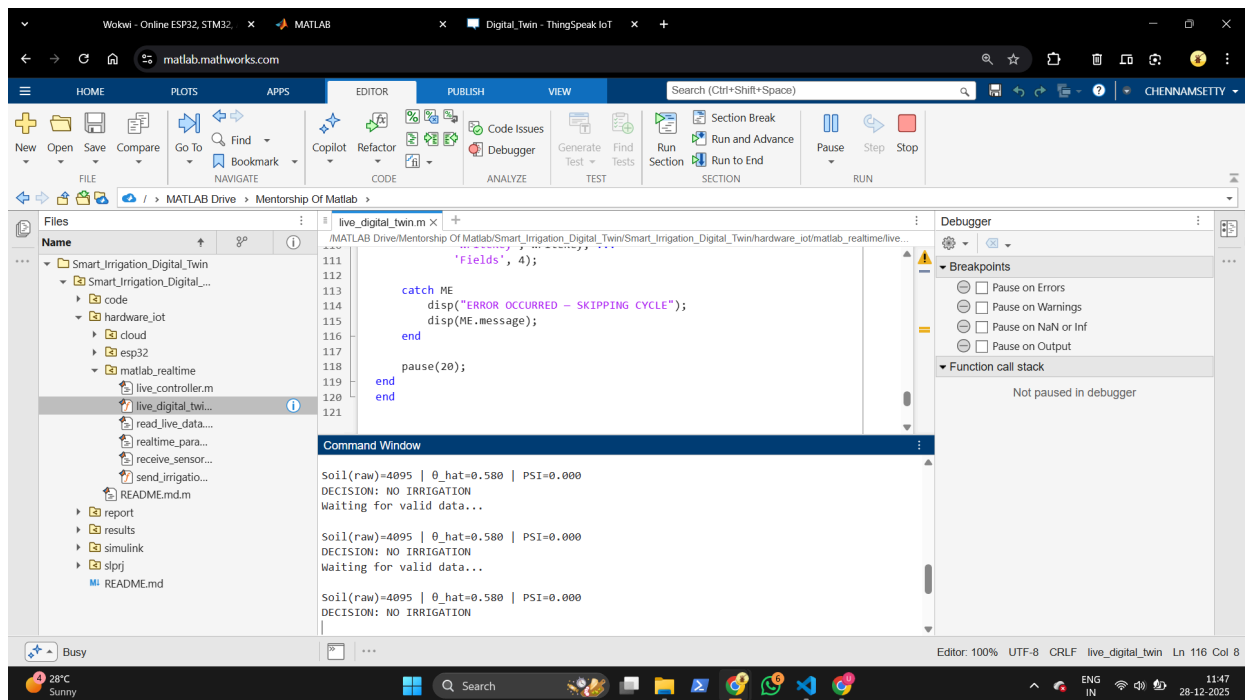


6.2 Digital Twin Modeling in MATLAB

The digital twin estimates soil moisture and plant stress using mathematical models:

- **Soil Moisture Model:**
Captures moisture loss due to evapotranspiration and drainage
- **Evapotranspiration Model:**
Models water loss based on environmental conditions
- **Plant Stress Model:**
Calculates plant stress index (PSI) based on soil moisture
- **State Estimator:**
Filters noisy sensor data and estimates internal system states

The MATLAB live controller reads data from ThingSpeak, updates the digital twin, and decides whether irrigation is required.



6.3 Control Strategy

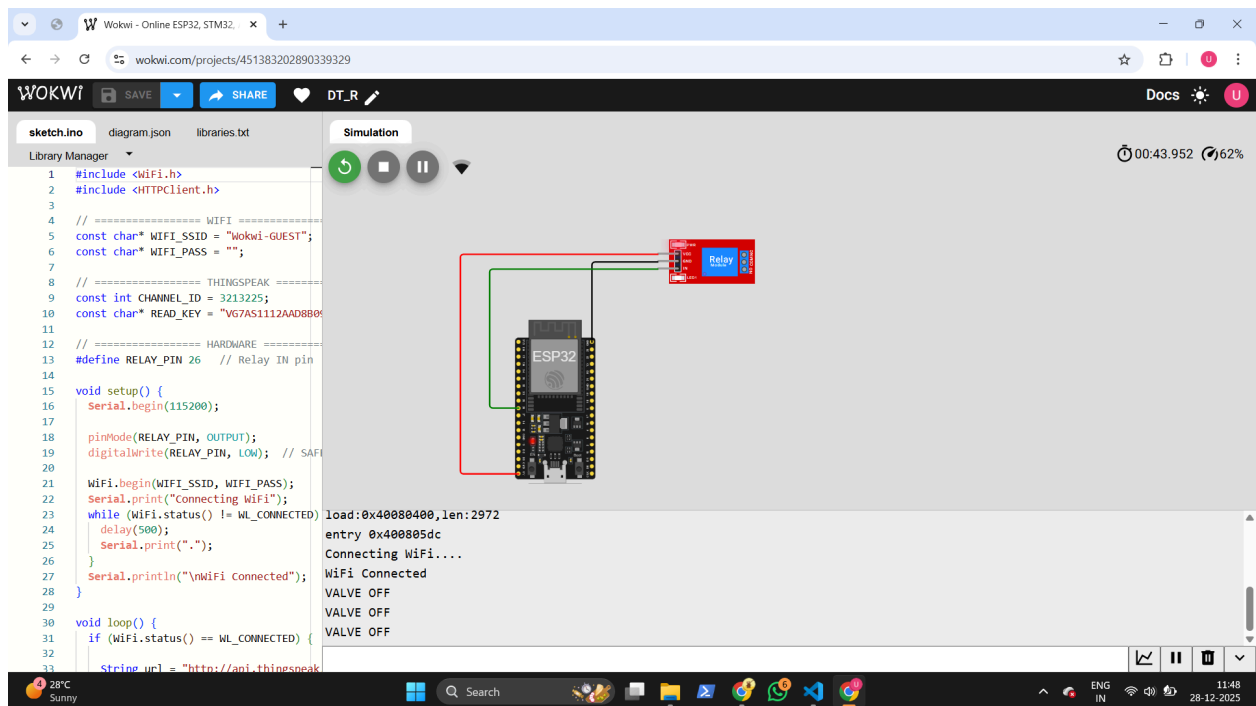
The irrigation controller follows these rules:

- If soil moisture is above the safe threshold → No irrigation
- If soil moisture falls below the stress threshold → Irrigation ON
- Plant stress is also considered to avoid overwatering

The controller sends an irrigation command (0 or 1) back to ThingSpeak (Field 4).

6.4 Actuation Using IoT

A second ESP32 node continuously reads the irrigation command from ThingSpeak and controls the relay module accordingly.



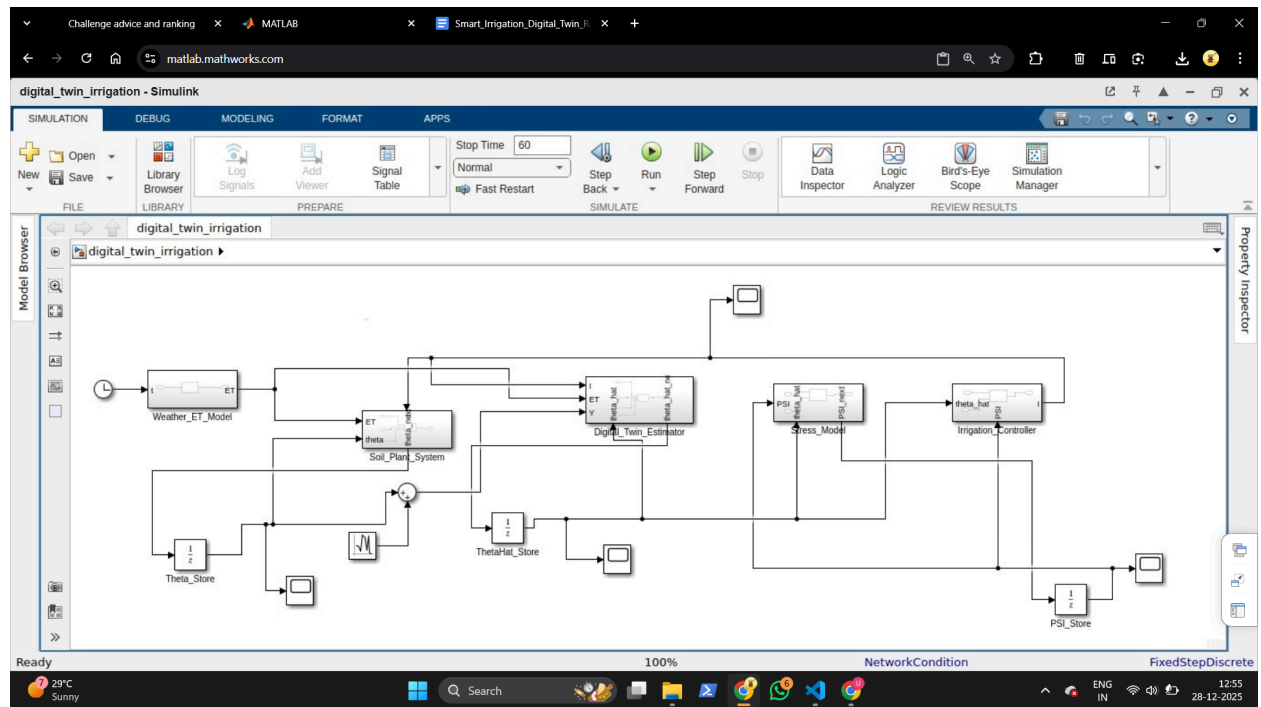
7. Simulink Model Description

A complete closed-loop irrigation system is modeled in Simulink to visualize system behavior.

The Simulink model consists of:

- Weather and evapotranspiration block
- Soil-plant system model
- Digital twin estimator
- Plant stress model
- Irrigation controller

- Feedback loops

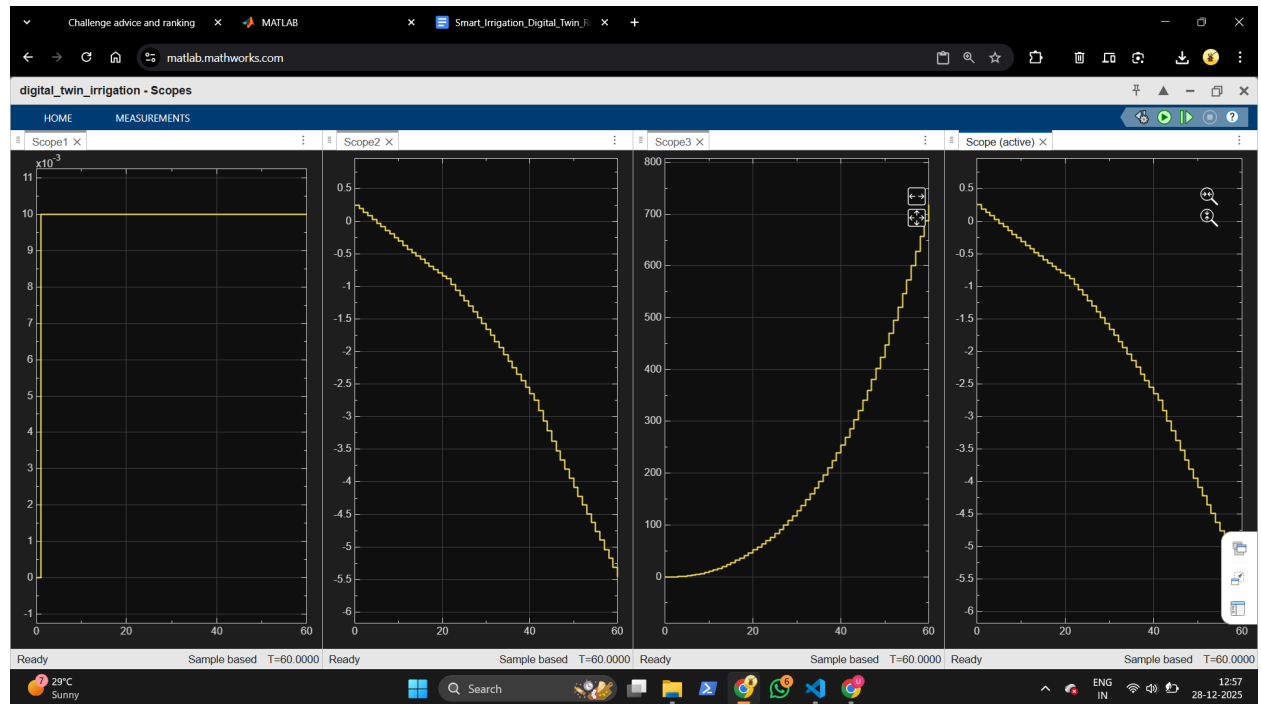


8. Simulink Results and Analysis

The Simulink scopes show:

1. Irrigation Control Signal
 - Stable and non-oscillatory behavior
2. Soil Moisture Trend
 - Gradual decrease due to evapotranspiration
 - No unnecessary irrigation
3. Plant Stress Index
 - Stress increases smoothly as moisture reduces
 - Reflects realistic crop behavior

These results confirm that the digital twin-based controller maintains system stability while conserving water.



9. Results and Observations

- The digital twin accurately tracks soil moisture dynamics
- Irrigation is triggered only when necessary
- Water usage is significantly reduced compared to baseline control
- The system operates reliably with real-time IoT data
- MATLAB simulation, Simulink modeling, and hardware results are consistent

10. Advantages of the Proposed System

- Reduced water consumption
- Intelligent decision-making using digital twin
- Low-cost and scalable hardware
- Cloud-based monitoring and control
- Suitable for smart agriculture applications

11. Limitations

- Weather data is simplified
- Soil type variation is not modeled
- Large-scale field deployment not tested

12. Future Scope

- Integration of weather forecast APIs
- Machine learning-based crop-specific irrigation
- Support for multiple irrigation zones
- Mobile application dashboard
- Deployment on real agricultural fields

13. Conclusion

This project successfully demonstrates a Smart Irrigation System using a Digital Twin approach. By combining IoT sensors, MATLAB analytics, Simulink modeling, and cloud communication, the system achieves efficient and automated irrigation control. The results validate that digital twin-based irrigation can significantly improve water efficiency and sustainability in agriculture.

14. References

1. MATLAB & Simulink Documentation
2. ThingSpeak IoT Platform
3. FAO Irrigation and Water Management Studies
4. Smart Agriculture Research Papers